

# INTRODUCTION TO ALGEBRA

## HISTORY OF ALGEBRA

Algebra is the invention of muslims. In 820 AD, a muslim mathematician Muhammad Ibne Mosa Al Khwarzmi wrote a book named Algebra-wal-Almuqabla. The first translation of this book named "Algebra" was published in Europe in Latin which is the base of the subject "Algebra".



## 8.1 ALGEBRA

**Explain the term algebra as an extension of arithmetic in which letter replace the numbers**

Man has been using natural numbers 1, 2, 3, ... in his life. As time passed his needs increased and the calculation became more complex. Algebra provides methods to transform and solve difficult and complex mathematical problems in easy way.

In Arithmetic we usually perform calculation of numbers with the help of fundamental operations addition, subtraction, multiplication and division, for example Rafay has 5 apples and Taha has 8 apples. So, the total number of apples is  $5 + 8$ .

The term algebra is an extension of arithmetic in which letters or symbols replace the numbers or quantities. For example Bisma has 5 toffees in a plate and Hifza has some toffees in her purse.

We can represent the total toffees as  $5 + x$ .

In this example we have shown the unknown quantity of toffees of Hifza by " $x$ ".



### Teacher's Note

*Teacher should give some more examples of unknown quantities from class room or daily life and represent them by  $x, y$  or  $z$ . Also add or subtract them with known quantities.*

Hence 'the sum of 5 and  $x$ ' is written in symbols as ' $5 + x$ '

Similarly other expression can be written in symbols as explained in the example.

**Example:** Represent into symbols.

| In words                               | In symbols<br>(Algebraic expression) |
|----------------------------------------|--------------------------------------|
| Sum of 6 and $x$                       | $6 + x$                              |
| Three times a number $y$               | $3y$                                 |
| The product of two numbers $x$ and $y$ | $xy$                                 |
| Five is greater than four              | $5 > 4$                              |
| Double of a number $e$                 | $2e$                                 |

**Know that a sentence is a set of words making a complete grammatical structure and conveying full meaning**

We speak sentences to convey some ideas in our daily life.

For example: Sum of two and seven is nine.

Symbolically, we write as  $2 + 7 = 9$

This sentence conveys full meaning.

So, In languages a sentence is set of words making a complete grammatical structure and conveying full meaning.

**Example:** Represent the following sentences symbolically.

(1) Double of a number  $x$  is 8

**Solution:** Symbolically we represent it as  $2x = 8$ .

(2) Five is greater than three

**Solution:** Symbolically we represent it as  $5 > 3$ .

**Know that Sentences that are either true or false are known as statements and a statement must be either true or false but not both information**

We, sometimes speak true sentences or false sentence for example.

(1) Difference of nine and two is seven

i.e.  $9 - 2 = 7$

It is a true sentence.

(2) Sum of two and five is four

i.e.  $2 + 5 = 4$

It is a false sentence.

Such sentences are known as statements.

Thus

Sentences that are either true or false are known as statements.

**A sentence that does not include enough information required to decide whether it is true or false is known as open statement (e.g.  $\Delta + 2 = 9$ ) and a number that makes an open statement true is said to satisfy the statement (e.g.  $\Delta = 7$  makes the statement  $\Delta + 2 = 9$  true). Also use English alphabet  $x$  in the open statement  $\Delta + 2 = 9$  to modify it to  $x + 2 = 9$**

We know that a statement must be either true or false. But in some cases we have to decide whether a statement is true or false. Such statement is called **open statement**.

**For example:**  $\Delta + 5 = 12$  is an open statement.

if we replace  $\Delta$  with 4 it becomes false i.e.  $4 + 5 = 12$  but if we replace  $\Delta$  with 7 it becomes true. i.e.  $7 + 5 = 12$ .

If  $\Delta = 2$ ,  $2 + 5 = 7$ , which is not equal to 12. So, it becomes false.

So, we say 7 satisfies the statement

and if we replace  $\Delta$  with English alphabet  $x$  it still remains open i.e.  $x + 5 = 12$

**Example 1:** Think and write the value of  $\Delta$  which makes the following statements true.

(i)  $\Delta + 6 = 18$                       (ii)  $\Delta - 3 = 7$

**Solution:** (i)  $\Delta + 6 = 18$

It will be true if  $\Delta = 12$  i.e.  $12 + 6 = 18$

(ii)  $\Delta - 3 = 7$

It will be true if  $\Delta = 10$  i.e.  $10 - 3 = 7$



**Example 2:** Replace the unknowns by the numbers to make statement true.

(i)  $2 + \square = 8$                       (ii)  $17 - \Delta = 9$

(iii)  $10 \div O = 5$                       (iv)  $6 \times x = 42$

**Solutions:**

(i)  $2 + \square = 8$

It will be true if  $\square = 6$  i.e.  $2 + 6 = 8$

(ii)  $17 - \Delta = 9$

It will be true if  $\Delta = 8$  i.e.  $17 - 8 = 9$

(iii)  $10 \div O = 5$

It will be true if  $O = 2$  i.e.  $10 \div 2 = 5$

(iv)  $6 \times x = 42$

It will be true if  $x = 7$  i.e.  $6 \times 7 = 42$

## EXERCISE 8.1

1. Write the following sentences in symbols.

- (i) Sum of a number  $x$  and two is eight.
- (ii) Product of a number  $y$  and seven is greater than two.
- (iii) Sum of six and a number  $z$  is less than four.

2. Which of the following are true or false statements?

- (i)  $5 + 9 = 18$       (ii)  $8 + 3 = 11$       (iii)  $20 + 3 = 26$
- (iv)  $21 - 8 = 13$       (v)  $18 - 13 = 15$       (vi)  $16 - 10 = 6$

3. Which of the following are open statements?

- (i)  $\Delta + 3 = 9$       (ii)  $3a + 4 = 8$       (iii)  $9 + 4 = 13$

4. Find the values of the number  $\Delta$  or  $\square$  which makes the following statements true.

- (i)  $6 + \Delta = 15$       (ii)  $\square + 5 = 8$       (iii)  $14 - \Delta = 8$
- (iv)  $24 - \Delta = 20$       (v)  $3\square + 1 = 10$       (vi)  $35 - \square = 29$

**Define variables as letters used to denote numbers in algebra**

Consider an open statement  $x + 5 > 2$

It will be true for different values of number  $x$ .

**For example:**

for  $x = 1$  we get  $1 + 5 > 2$  or  $6 > 2$

for  $x = 2$  we get  $2 + 5 > 2$  or  $7 > 2$

for  $x = 3$  we get  $3 + 5 > 2$  or  $8 > 2$

and so on

It means the value of number  $x$  is not constant, but it is varying.

So,  $x$  is called variable.

Hence

Any of the letter of English alphabet which is used to denote number in algebra is called variable.

**Know that any numeral, variable or combination of numerals and variables connected by one or more of the symbols '+' and '-' is known as an algebraic expression (e.g.,  $x + 2y$ )**

In daily life, we often speak set of words which are not sentences.

**For example:**

(1) Difference of a number  $x$  and two

Symbolically we write as  $x - 2$

(2) Sum of thrice of a number  $x$  and twice of a number  $y$

Symbolically we write as  $3x + 2y$

Such expressions like  $x - 2$  and  $3x + 2y$  are known as algebraic expression. In these expressions we see variables, numerals and fundamental operations.

Thus

Algebraic expression is the combination of variables, numerals and fundamental operations.

$2 + x$ ,  $5x$ ,  $6x - 3y + 8$  are few examples of algebraic expressions.

**Example 1:**

Moomal has three brothers. Every brother gives  $x$  rupees to Moomal. She already has Rs 7.

- (i) Write an algebraic expression for the amount she has.
- (ii) If each brother gives Rs 10. Find the amount that Moomal has.
- (iii) If each brother gives him Rs 15. Find the amount that she has.

**Solution:**

- (i) Three brothers give her  $x + x + x = 3x$   
Already Moomal has Rs 7  
Total amount that Moomal has  $3x + 7$
- (ii) If  $x = 10$ , then Moomal has  
 $3x + 7 = 3 \times 10 + 7 = 37$  rupees
- (iii) If  $x = 15$ , then Moomal has  
 $3x + 7 = 3 \times 15 + 7 = 52$  rupees

## 8.2 ALGEBRAIC EXPRESSIONS

**Know that  $x$ ,  $2y$  and  $5$  are called the terms of the expression  $x + 2y + 5$**

The parts of an algebraic expressions are connected to each other by the signs of addition and subtraction, for example in  $5x + 4y$ , there are two parts,  $5x$  is 1st part and  $4y$  is 2nd part. They are connected by plus sign. The parts  $5x$  and  $4y$  are called the 'terms' of the expression  $5x + 4y$ .

Similarly  $a+b$  is an algebraic expression having two terms i.e.  $a$  and  $b$ .

The expressions  $a \div b$  or  $\frac{a}{b}$  and  $ab$ , contain only one term.

An algebraic expression consists of terms which are connected by the operations of addition (+) and subtraction (-).

**Note:** Operations of multiplication ( $\times$ ) and division ( $\div$ ) do not connect terms.



**Example:** Write the number of terms in the following expressions:

- (1)  $5x + 7y$       (2)  $6x \div 7 + y$       (3)  $3x - 4y - 7$       (4)  $4 \times p \times q$

**Solutions:**

(1)  $5x + 7y$

Number of terms = 2

(2)  $6x \div 7 + y$

Number of terms = 2

(3)  $3x - 4y - 7$

Number of terms = 3

(4)  $4 \times p \times q$

Number of terms = 1

**Know that the symbol or number appearing as multiple of a variable used in algebraic term is called its coefficient (e.g. in  $2y$ , 2 is the coefficient of  $y$ )**

In algebraic expression  $3x + 7$ , the term  $3x$  contains a variable  $x$  and a number 3 which is multiplied with  $x$ . In this term 3 is the coefficient of  $x$ .

In algebraic expression, the symbol or number appearing as multiple of a variable used in algebraic term is called its coefficient. e.g. in  $4x$ , 4 is the coefficient of  $x$ .

Similarly, in algebraic expression  $4a - 3c$ , 4 is the coefficient of  $a$  and  $-3$  is the coefficient of  $c$ .

Similarly, Coefficient of  $x$  is 1, which is not written.  
and the coefficient of  $-x$  is  $-1$ , which is also not written.

*Generally  $x$ ,  $y$  and  $z$  are used to write variable and  $a$ ,  $b$ ,  $c$  are used to denote numbers.*

**Know that the number, appearing in algebraic expression, independent of a variable is called a constant term (e.g. in  $x + 2y + 5$ , number 5 is a constant term).**

In the algebraic expression  $3x + 7$  if  $x = 10$ , the value of  $3x$  is 30 and for  $x = 15$ , the value of  $3x$  is 45. In both cases the second term 7 of the expression  $3x + 7$  does not change because 7 is a constant. Similarly in  $2x + 9$ , 9 is constant.

In algebraic expression number appearing independent of a variable is called constant term.

**Example:** Write coefficients, constants and variables of the following expressions.

(1)  $6z - 8$

(2)  $2x + 3y + 8$

**Solution:**

(1)  $6z - 8$

Here Coefficient is: 6

Constant is:  $-8$

Variables are:  $z$

(2)  $2x + 3y + 8$

Here Coefficients are: 2, 3

Constant is: 8

Variable is :  $x, y$

### Differentiate between like and unlike terms

The terms having same variable are called 'like terms'. In like terms coefficients and their signs may differ.

For example,  $2x$ ,  $-3x$  and  $\frac{5}{2}x$  are like terms.

The terms having different variables are called 'unlike terms'. For example  $3x$  and  $4y$  are unlike terms. Similarly  $-3t$  and  $5m$  are unlike terms.

Like terms can be combined to give a single term.

As  $2x + 3x = 5x$

Here two like terms  $2x$  and  $3x$  have been combined to give a single term  $5x$ .

So,

Addition or subtraction can not be performed with unlike term

As unlike term  $5x$  and  $7y$  can not be combined to give single term i.e.  $5x + 7y$  can not be further simplified.



## EXERCISE 8.2

## 1. Write the number of terms of the following expressions.

- (i)  $3 \times m \times n$       (ii)  $p$       (iii)  $2x + 3y$   
(iv)  $4y + 7$       (v)  $4l + 3m + 9$       (vi)  $7x + 5 + 8$

## 2. Write the variables of the following expressions.

- (i)  $8x + 3$       (ii)  $-7x + 1$       (iii)  $6x + y$       (iv)  $\frac{2x}{y}$   
(v)  $2xy$       (vi)  $5 \div 3yz$       (vii)  $3xyz$       (viii)  $3xy + 9yz$

## 3. Write the constants of the following expressions.

- (i)  $2x + 2$       (ii)  $-3 + x$       (iii)  $xy + 1$   
(iv)  $xy + yz + \frac{1}{2}$       (v)  $5x + 9$       (vi)  $2x + 2y + 10$

## 4. Write the coefficients of variables of the following.

- (i)  $-7x$       (ii)  $5y$       (iii)  $2x + 3y$   
(iv)  $x + \frac{1}{2}y - \frac{1}{4}z$       (v)  $5x + 9$       (vi)  $\frac{1}{4}x + 6y + 1$

## 5. Write like terms in the following.

$2xy, 4lm, -7xz, 140xy, 13mn, xz, \frac{1}{5}xy, -9xyz, \frac{1}{4}xy, -2p,$   
 $\frac{1}{3}xy, -2xy, 5xy, xyz, 46p, 5lm$

## Add and subtract given algebraic expressions

## 1. Addition of algebraic expressions

We know that  $x + x = 2x$  and  $x + x + x = 3x$

Here coefficient shows how many times the variable is added.

$$2x + 3x = (x + x) + (x + x + x)$$

$$= x + x + x + x + x = 5x$$

Thus, to add like terms only their coefficients are added and bases and exponents remain the same.

Let us see some examples.

**Example 1:** Add  $7x$  and  $5x$

**Solution:**

*Writing the terms vertically:*

$$\begin{array}{r} 7x \\ + 5x \\ \hline 12x \end{array}$$

*Writing the terms horizontally:*

$$7x + 5x = (7 + 5)x = 12x$$

It should be noted that in the addition of  $7x$  and  $5x$  only coefficients are added and variable  $x$  remains the same.

**Example 2:** Find the sum of  $3a + 2b + c$  and  $8c + 6b + a$

**Solution:**

*Writing the terms vertically:*

$$\begin{array}{r} + 3a + 2b + c \\ + a + 6b + 8c \\ \hline 4a + 8b + 9c \end{array}$$

*Writing the terms horizontally:*

$$\begin{aligned} \text{Sum} &= (3a + 2b + c) + (a + 6b + 8c) \\ &= 3a + 2b + c + a + 6b + 8c \\ &= 3a + a + 2b + 6b + c + 8c \\ &= 4a + 8b + 9c \end{aligned}$$

**Example 3:** Add the following:

**Solution:**  $x + 2xy + y, 3x + 5xy, 5x + xy + 7y$

$$\begin{array}{r} x + 2xy + y \\ 3x + 5xy \\ 5x + xy + 7y \\ \hline 9x + 8xy + 8y \end{array}$$

- While adding vertically like terms are arranged in same column and then added.
- While adding horizontally like terms are arranged together and added.

**2. Subtraction of Algebraic Expressions**

We know that subtraction is the inverse operation of addition.  
For example to subtract 3 from 4 is equivalent to “add - 3 in 4”.

$$\begin{aligned} (+4) - (+3) &= (+4) + (-3) \\ &= 1 \end{aligned}$$

So to subtract a number from a given number simply change the sign of subtrahend (from - to + and + to -) and add it in the given number.

**Example 1:** Subtract  $6x$  from  $9x$ .

**Solution:**

*Writing the terms horizontally:*

$$\begin{aligned} \text{Difference} &= (+9x) - (+6x) = (+9x) + (-6x) \\ &= 9x - 6x = 3x \end{aligned}$$

*Writing the terms vertically:*

$$\begin{array}{r} 9x \\ + 6x \\ \hline 3x \end{array}$$

**Example 2:** Subtract  $-6y$  from  $10y$ .

**Solution:**

*Writing the terms horizontally:*

$$\begin{aligned} \text{Difference} &= (+10y) - (-6y) = (+10y) + (+6y) \\ &= 10y + 6y = 16y \end{aligned}$$

*Writing the terms vertically:*

$$\begin{array}{r} 10y \\ + 6y \\ \hline 16y \end{array}$$

**Example 3:** Subtract  $3x - 2y$  from  $8x + 4y$

**Solution:**  $(8x + 4y) - (3x - 2y) = 8x + 4y - 3x + 2y$

$$\begin{aligned} &= (8x - 3x) + (4y + 2y) \\ &= 5x + 6y \end{aligned}$$

This question can also be solved as under:

$$\begin{array}{r} + 8x + 4y \\ + 3x + 2y \quad (\text{by changing signs}) \\ \hline 5x + 6y \end{array}$$



**Example 4:** Subtract  $5x + 2y$  from  $8x + 5y$ **Solution:**

$$\begin{array}{r}
 + 8x + 5y \\
 \pm 5x \pm 2y \quad (\text{by changing signs}) \\
 \hline
 3x + 3y
 \end{array}$$

**EXERCISE 8.3****1. Add the following expression.**

(i)  $2x, x, 4x$

(ii)  $2a, 3a, 6a, a$

(iii)  $8lm, 4lm, lm, 6lm$

(iv)  $2xy, 4xy, xy, 6xy, 3xy$

**2. Simplify:**

(i)  $2x + 9x$

(ii)  $a + 2a + 3a$

(iii)  $x + 3x + 6x + 10x$

(iv)  $2st + 3st + 5st + 7st$

**3. Add:**

(i)  $3a + 2b, a + b, 4a$

(ii)  $6x + 5y + 7z, 2x + 3y + z, y + 2x$

(iii)  $pq + qr + pr, qr + 4pr + 2pq, 3pq + 2pr$

(iv)  $3c + 4d + 5f, 5c + 7d + 6f$

**4. Subtract**

(i)  $5x$  from  $8x$

(ii)  $-3y$  from  $9y$

(iii)  $2x + 3y$  from  $6x + 8y$

(iv) Subtract  $-20f + 30g + 40$  from  $20 + 10f + 20g$ .

**5. Perform the subtraction.**

(i) 
$$\begin{array}{r}
 8x \\
 + 5x \\
 \hline
 \hline
 \end{array}$$

(ii) 
$$\begin{array}{r}
 12ab \\
 - 9ab \\
 \hline
 \hline
 \end{array}$$

(iii) 
$$\begin{array}{r}
 5z \\
 + 3z \\
 \hline
 \hline
 \end{array}$$

(iv) 
$$\begin{array}{r}
 10xy \\
 - 17xy \\
 \hline
 \hline
 \end{array}$$

(v) 
$$\begin{array}{r}
 2x + 7y \\
 - 3x + 5y \\
 \hline
 \hline
 \end{array}$$

(vi) 
$$\begin{array}{r}
 2x + 15xb + 9y \\
 + x + 20xb + 3y \\
 \hline
 \hline
 \end{array}$$

**Simplify algebraic expressions grouped with brackets**

We know that brackets are used to indicate the order for performing operations. The four kinds of brackets are:

- (i) “—” is called a bar or vinculum.
- (ii) “( )” is called a round or curved brackets or parentheses.
- (iii) “{ }” is called a curly brackets or braces.
- (iv) “[ ]” is called box brackets or square brackets.

In algebra, sometimes, we can't simplify an expression into a single term within the brackets. For example, in  $2x - (x + y)$  we can't simplify the expression  $(x + y)$ . For such a situation.

- (i) Expand the brackets
- (ii) Simplify the whole expression as given below:  

$$2x - (x + y) = 2x - x - y = x - y$$

Hence,  $x - y$  is the simplest form of the above mentioned algebraic expression.

-ve sign before the brackets means, change the signs of all the terms within brackets, i.e.  $-(a + b) = -a - b$ .

**Example 1:** Simplify the following.

- (i)  $[5a - \{3b + (6a - 2a + b)\}]$
- (ii)  $[2a + \{c - a + (a + 2b + c)\}]$
- (iii)  $xy - [yz - \{zx + xy + 9yz - \overline{zx + xy}\}]$

**Solutions:**

$$\begin{aligned}
 \text{(i)} \quad & [5a - \{3b + (6a - 2a + b)\}] \\
 &= [5a - \{3b + (4a + b)\}] \\
 &= [5a - \{3b + 4a + b\}] \\
 &= [5a - \{4a + 4b\}] \\
 &= [5a - 4a - 4b] \\
 &= [a - 4b] = a - 4b
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad & [2a + \{c - a + (a + 2b + c)\}] \\
 &= [2a + \{c - a + (a + 2b + c)\}] \\
 &= [2a + \{c - a + a + 2b + c\}] \\
 &= [2a + \{2b + 2c\}] \\
 &= [2a + 2b + 2c] \\
 &= 2a + 2b + 2c
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad & xy - [yz - \{zx + xy + (yz - zx + xy)\}] \\
 &= xy - [yz - \{zx + xy + (yz - zx - xy)\}] \\
 &= xy - [yz - \{zx + \cancel{xy} + yz - zx - \cancel{xy}\}] \\
 &= xy - [yz - yz] = xy - [yz - yz] = xy - 0 = xy
 \end{aligned}$$

**Example 2:**

Simplify the following expression

$$25 - [-7a - \{-6a + (3 - 5 - 6a)\}]$$

**Solution:**

$$\begin{aligned}
 &= 25 - [-7a - \{-6a + (3 - 5 - 6a)\}] \\
 &= 25 - [-7a - \{-6a + (-2 + 6a)\}] \\
 &= 25 - [-7a - \{-6a - 2 + 6a\}] \\
 &= 25 - [-7a - \{-2\}] \\
 &= 25 - [-7a + 2] \\
 &= 25 + 7a - 2 \\
 &= 7a + 23
 \end{aligned}$$

**Evaluate and simplify an algebraic expression when the values of variables involved are given**

The process of finding the value of an expression by using numbers in place of variables is called evaluation.

Some examples of evaluation are given as under.

**Example:** If  $a = 2$ ,  $b = -3$  and  $c = -4$ , then evaluate the following.

(i)  $ab + bc + ca$

(ii)  $\frac{bc(b-c)}{a}$

**Solution:**

(i)  $ab + bc + ca$

$$\begin{aligned}
 &= 2 \times (-3) + (-3) \times (-4) + (-4) \times 2 \\
 &= -6 + 12 - 8 \\
 &= -14 + 12 = -2
 \end{aligned}$$

(ii)  $\frac{bc(b-c)}{a}$

$$\begin{aligned}
 &= \frac{(-3) \times (-4) [(-3) - (-4)]}{2} \\
 &= \frac{12 [-3 + 4]}{2} = \frac{12 \times 1}{2} = 6
 \end{aligned}$$



## EXERCISE 8.4

## 1. Simplify the following expressions.

(i)  $[x + x + (y + y + 2x)]$

(ii)  $[5a - \{3a + (3b - 6b + 4a)\}]$

(iii)  $[2a - \{5b - 3(2a + b - 3a)\}]$

(iv)  $5\ell - [2 - (3 + 11\ell + 3m + 8\ell)]$

(v)  $2(x + y) - 3[4x + \{7x + 3y + (5x - 4y)\}]$

2. (i) If  $x = 2$ , find the values of  $3x$  and  $-5x$ (ii) If  $y = -3$ , find the values of

(a)  $-3y$  (b)  $5y + 7$  (iii)  $\frac{5}{3}y$  (iv)  $\frac{6}{24}y + 9$

3. If  $a = -3$ ,  $b = 5$  and  $c = -2$ , evaluate the following:

(i)  $a + b + c$

(ii)  $2a - 3b + c$

(iii)  $\frac{a - bc}{c}$

(iv)  $\frac{2a + b + c}{abc}$

(v)  $\frac{a + b}{c} + \frac{b + c}{a}$

(vi)  $\frac{2a + b - c}{a - 3b + c}$

(vii)  $\frac{a(2b + 3c)}{3ca}$

(viii)  $a + 3abc - b + c$

(ix)  $\frac{3a + 4b + 5c}{a + b - c}$

(x)  $\frac{a}{b} - \frac{b}{c} + \frac{c}{a} - \frac{a}{c}$

4. If  $x = 2$ ,  $y = 3$  then evaluate the following expression.

$8x + [3y - \{6x + (5 - 4x)\}]$

5. Abid purchased 5 bags at the rate of Rs  $x$  per bag. After paying he has left Rs 3.

Then,

(i) Find the cost of 5 bags.

(ii) Write an algebraic expression for paying the cost of 5 bags and Rs 3 left with him.

(iii) If  $x = 30$ , find the total amount that Abid had.

## REVIEW EXERCISE 8

1. Write in symbols.
  - (i) Cost of  $x$  books, when price of one book is Rs 18.
  - (ii)  $9b$  plus  $5a$
  - (iii) 3 times  $x$  plus 5
  - (iv) Twice of  $y$  minus two third
  - (v) 6 more than  $p$
2. Write the following in words.
  - (i)  $x + 2$
  - (ii)  $3y - 4$
  - (iii)  $x + 5$
  - (iv)  $s + 2t$
  - (v)  $\frac{x - y}{2}$
3. Write the number of terms in each of the following expression.
  - (i)  $2a + 5b$
  - (ii)  $16x$
  - (iii)  $2bc + 2a - c$
  - (iv)  $3xy - 8x + 6y - 2$
  - (v)  $6x \div 5y + 2z - 6$
4. Write three expressions containing one term, two terms and three terms respectively.
5. Simplify:
  - (i)  $4xy + 2xy + \frac{2}{3}xy$
  - (ii)  $9ab + 3ab + \frac{3}{5}ab$
6. Add:
  - (i)  $3x + 4y$ ,  $5x + 9y$
  - (ii)  $2d + 5c + 3b$ ,  $7c + 5d + 9b$
  - (iii)  $12xy + 3x + 4y$ ,  $5x + 6y + 8xy$
7. Subtract  $x + 3y$  from  $2x + 7y$
8. Subtract  $x + 3y + 5z$  from  $2x - 15y - 9z$
9. Which of the following are true or false statements?
  - (i)  $13 - 5 = 3$  ( )
  - (ii)  $13 - 5 = 5$  ( )
  - (iii)  $13 - 5 = 8$  ( )
10. Find the value of variable which makes the statement true.
  - (i)  $x = 15 - 6$
  - (ii)  $y - 8 = 5$
  - (iii)  $14 = b + 8$
  - (iv)  $4x - 10 = 2$
  - (v)  $12 - 2m = 8$

**SUMMARY**

- Algebra is generalized arithmetic in which letters (called variables) are also used as numbers.
- A sentence is a set of words making a complete grammatical structure and conveying full meaning.
- A sentence is called TRUE if it satisfies the given condition and FALSE if it does not satisfy the condition.
- A statement is called OPEN if it is either true or false.
- An algebraic expression is the combination of variables, numerals and fundamental operation.
- The number appearing before a variable in a term of algebraic expression is called its **coefficient**.
- The terms having the same variables are called **like terms** otherwise, they are **unlike terms**.
- To add two (or more) algebraic expressions, add their like terms according to the rules for the addition of integers.
- To subtract one expression from the other, change the sign of each term of the expression to be subtracted and add to the expression from which the subtraction is to be performed.
- If an algebraic expression involves two or more than two operations, it is simplified following the rules of BODMAS.
- To evaluate any algebraic expression means to find its value after substituting the given numerical values of each variable in the expression.